COINjecture Network
A Mathematics-Backed Peer-to-Peer Network

COINjecture.com

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Abstract

COINjecture is a purely peer-to-peer network that harnesses the computational asymmetry benefits inherent to NP-complete mathematical problems to replace the traditional proof-of-work hashing mechanism needed to outpace attackers and secure a robust network of digital payments. Chain validity derives from solution correctness and cumulative mathematical work, ensuring robust consensus without centralized validators. The record formed in the process of solving and verifying these NP-complete problems creates a computational marketplace where miners earn rewards for solving computationally hard problems while enabling secure, decentralized transactions.

Code Base

- **Active Testnet**: https://github.com/beanapologist/coinjecture
- **Archive**: https://gitlab.com/beanapologist/coinjecture
- **Archive**: https://codeberg.org/beanapologist/COINjecture

1 Introduction

Traditional proof-of-work blockchain systems, like Bitcoin, rely on the Secure Hash Algorithm 256-bit (SHA-256) as the mechanism to verify transactions, link blocks, and maintain network integrity and chain immutability. The SHA-256 algorithm is applied to hash a block header repeatedly until one is found with a specific prefix of leading zeros. During this process a random number called a nonce is changed over and over with each attempt in a brute-force effort to find the solution. This hash calculation provides no scientific or mathematical value outside of network security and the computational resources required to perform this brute-force search resulted in the consumption of over 150 TWh in 2024, according to the Cambridge Bitcoin Electricity Consumption Index (CBECI). This computation provides network security but no additional utility.

What is needed is a cryptographically secure electronic payment system that generates utility beyond security. Building on Bitcoin's foundation of decentralized consensus through proof-of-work, COINjecture proposes to replace SHA-256 with NP-complete mathematical problem solving that records verifiable, immutable computational data on-chain. Consensus is derived from mathematical correctness and the overall work performed by the chain. The computational work that fuels chain growth and provides network consensus also generates economic utility to establish a computational marketplace in which miners are rewarded for solving computationally hard problems.

2 Transactions

COINjecture follows the Standard UTXO model in which we define an electronic coin as a chain of digital signatures. Each owner transfers the coin by digitally signing a hash of the previous transaction and the public key of the next owner. In addition, all blocks in our network must contain not just transactions, but also proofs of computational work (e.g., NP-hard problem solutions) that secure consensus and provide economic utility. This data alongside transactions are broadcast to all nodes in the peer network.

3 Timestamp Server

Like Bitcoin, we employ a timestamp server that proves that the data must have existed at the time of announcement. Each timestamp includes the previous timestamp in its hash as well as the computational proofs (problem + solution) of the block, forming a chain.

4 Proof-of-Work

To implement a distributed timestamp server on a peer-to-peer basis, we use a proof-of-work system based on NP-hard problems rather than hash collision. The defining feature of an NP problem is that a candidate solution can be verified as correct, but not necessarily solved, in polynomial time so that given a potential answer to the problem, you can check if it's correct quickly, even if finding the answer in the first place is difficult or time-consuming. For instance, a sudoku puzzle may take several minutes to solve, but can be checked for correctness quickly by ensuring no row or column contains two occurrences of the same number.

5 Commitment Protocol

To prevent grinding or generating many problems to find easy instances, we employ a succinct commit-mine-reveal protocol, in which:

1. Miner computes: `commitment = H(problem_params || salt || H(solution))`
2. Miner finds header with valid commitment and sufficient work score
3. Miner publishes solution bundle, and prove commitment matches

The epoch salt is derived from parent block hash to prevent pre-mining activities. Commitment cryptographically binds to solution via `H(solution)` while hash reveals nothing about the actual solution structure.

6 Problem Types

The commit-reveal protocol allows the network to use different NP-hard problem types.

7 Work Score Calculation

Unlike traditional, hash-based proof-of-work, where 'work' is primarily the number of hashes attempted (inferred from the difficulty target), our work score is a measurement of computational difficulty that takes into account solve and verification time, the correctness and optimality of the solution, and the size of the problem and its corresponding difficulty. The difficulty target adjusts based on recent work scores to maintain stable block time.

8 Network

The steps to run the network are as follows:

1. New transactions are broadcast to all nodes
2. Miners select problems (user-submitted or consensus-generated)
3. Miners solve NP-hard problems
4. Upon solving, miner create commitment and mines header
5. When valid header found, broadcast to all nodes
6. When commitment confirmed, broadcast solution reveal
7. Nodes validate solution and accept block if valid
8. Nodes express acceptance by working on next block using hash of accepted block

Nodes always consider the longest chain and the highest cumulative work to be correct and work on extending it. The cumulative work score is also the tie breaker mechanism in the event that two nodes broadcast different versions of the next block simultaneously. The asymmetry of NP-hard problems (fast verification, slow solving) means that nodes can quickly verify competing chains during a fork to determine which one is valid and has the most "weight" according to the tiebreaker rule.

9 Problem Pool

The network maintains two problem sources, consensus problems and user problems. Consensus problems include those that are generated deterministically from network state for base rewards. User problems are submitted with escrowed bounties.

10 Incentive

By convention, the first transaction in a block is a special transaction that starts a new coin owned by the creator of the block. This adds an incentive for nodes to support the network, and provides a way to initially distribute coins into circulation, since there is no central authority to issue them.

Unlike bitcoin, which uses the analogy of gold miners expending resources to add gold into circulation, COINjecture incentive is derived through the measured performance of the nodes completing the work, like a promotion based on the work you performed. In our case, it is empirical measured against the actual performances of the chain and includes the problem solution quality,

the solution speed, the time, and electricity that is expended. This quantifies "good work" and rewards it proportionally.

COINjecture does not utilize predetermined fee schedules, instead incentive emerges from the dynamics of the decentralized network state itself. We model the network as a coupled oscillator system where nodes synchronize through proof-of-work. All blockchain measurements must satisfy three fundamental principles:

1. **Empirical** - Derived from actual network behavior, not assumptions
2. **Self-referential** - Measured against the network's own state, not external units
3. **Dimensionless** - Pure ratios with no arbitrary limits or parameters

These principles ensure the system adapts naturally to its own evolution without manual intervention. The incentive may help encourage nodes to stay honest. If a greedy attacker is able to recalculate all of the correct solutions and produce the same quality of work as honest nodes, they would have to choose between using it to defraud people by stealing back their payments, or using it to generate new coins. They ought to find it more profitable to play by the rules, such rules that favour them with more new coins than everyone else combined, than to undermine the system and the validity of their own wealth.

11 The Conjecture

COINjecture tests whether optimal decentralized consensus naturally converges to the critical constant $\eta = \lambda = 1/\sqrt{2}$. Every percentage in the tokenomics derives from this equilibrium. If the network sustains stable operation across diverse conditions, it provides empirical evidence for a fundamental constant in distributed systems—analogous to how physical constants emerge from natural law.

The steady addition of useful computational work to the chain serves as both security mechanism and economic utility. Nodes are incentivized to perform and verify real mathematical work, creating value beyond network security.

12 Reclaiming Disk Space

Once the latest block in a chain is buried under enough blocks, solutions to problems can be discarded by mining nodes to save disk space. To facilitate this without breaking the block's hash, solutions and transactions are hashed in a Merkle tree. Block headers form the Merkle tree, with only the root included in the block's hash. Old blocks can be compacted by stubbing off branches of the tree.

A block header with no transactions or solutions is about 1 KB. If we suppose blocks are generated every 14.14 seconds ($\sim$0.24 minutes), storage per year is $\sim$ 2.4 GB per year. With computer servers typically selling with 32 GB of RAM as of 2025, storage should not be a problem even if the block headers must be kept in memory.

The network supports different node types with varying storage requirements:

- **Light Nodes**: Store only block headers and verify commitments. Can request full solutions from full nodes when needed. Suitable for mobile mining and everyday transactions.

- **Full Nodes**: Store complete blockchain and recent proof bundles (e.g., last N epochs). Validate all transactions and solutions. Can prune old solutions after sufficient confirmations.
- **Archive Nodes**: Maintain complete historical record of all proofs. Enable academic verification, reproducibility, and serve as reference for the network.

Mining can occur on any node type. Light miners request work from full nodes, solve problems locally, and submit solutions. This enables mining on resource-constrained devices while maintaining network security through full node validation. For user-submitted problems, solutions are recorded on-chain but full proof bundles can be pruned by non-archive nodes after the user retrieves their solution.

13 Privacy

Traditional privacy models rely on limiting access to information. COINjecture requires transparency of measurements while preserving user privacy of methods.

13.1 Public Information (Required for Verification)

The network must verify computational work, which requires:

- Problem parameters and solution
- Commitment timestamp (T_1) and reveal timestamp (T_2)
- Hardware class specification (for fair comparison)
- Work score measurements (solve_time, verify_time, memory, quality)

All nodes independently verify solution correctness and measure their own verification metrics. This transparency is necessary for trustless consensus.

Computational methods remain proprietary while measurements remain verifiable. This preserves both marketplace trust (public verification) and competitive advantage (private innovation).

Transaction privacy follows Bitcoin's model using pseudonymous addresses. The epoch salt, derived from the parent block hash, prevents transaction linking and ensures problems cannot be pre-computed before commitment. A new epoch salt should be used for each transaction to keep them from being linked by a common owner. This ensures that the network learns WHAT you solved (problem and solution), WHEN you solved it (T_1 and T_2), WHAT hardware you used (for fair comparison), and HOW LONG it took (solve_time = T_2 - T_1). The network never learns HOW you solved it (algorithm) or WHY your implementation is efficient (proprietary optimizations).

14 Calculations

Consider an attacker trying to generate an alternative chain faster than the honest network. The commit-reveal protocol prevents grinding (generating many problems to find easy ones): commitment binds to a specific problem before mining begins. Given this, the attacker must solve problems of similar difficulty to honest miners.

The probability they catch up from z blocks behind is p^z where $p = \text{attacker_rate}/\text{honest_rate}$. For $p < 0.5$, this drops exponentially. The asymmetry of NP-hard problems (exponential solve, polynomial verify) means nodes can quickly verify competing chains during forks to determine cumulative work. An attacker cannot present fake solutions because verification is fast.

15 Conclusion

We have proposed a blockchain system that replaces traditional SHA-256 proof-of-work mining with NP-hard mathematical computations to create a marketplace for novel computational data. Nodes work all at once with little coordination and the mathematical work performed by the network serves as the consensus mechanism. This creates the first blockchain where proof-of-work both secures the network and provides economic utility.

A Problem Registry

Our current testnet employs Subset Sum. Subset sum is a classic NP-hard problem with no known polynomial time algorithm. It asks if any subset of a given set of positive integers sums to a specific target value, denoted mathematically as:

Given a set of integers $S = \{s_1, s_2, \ldots, s_n\}$ and a target sum T, find a subset $S' \subseteq S$ such that $\sum_{s \in S'} s = T$.

Subset sum is a fundamental problem in computer science and often serves as the foundation for solving other NP-hard problems, making it ideal as our introductory problem type.

Other key NP-hard problems we intend to explore include:

- **Integer Factorization**: Given $N = p \times q$, find p and q. Complexity $L_n[1/3, c]$, verification $O(1)$ multiplication.
- **Graph Coloring**: Given a color for each vertex (node) of a graph, assign a color to each so that no two adjacent vertices have the same color. NP-complete, $O(V + E)$ verification.
- **Traveling Salesman Problem (TSP)**: Given cities and distances, find the shortest tour. NP-hard, $O(n)$ verification.
- **Shortest Vector Problem (SVP)**: Given a basis of a vector space V and a norm N for a lattice. NP-hard, $O(1)$ verification.
- **Boolean Satisfiability**: Given CNF formula, find satisfying assignment. NP-complete, $O(n)$ verification.

B Work Score Formula Breakdown

$$\text{work_score} = \text{time_asymmetry} \times \sqrt{\text{space_asymmetry}} \times \text{problem_weight} \times \text{size_factor} \times \text{quality_score} \times \text{Energy_efficiency} \tag{1}$$

B.1 Work Score Based Rewards

Rewards are calculated purely from:

- Time asymmetry: solve_time / verify_time
- Space asymmetry: solve_memory / verify_memory
- Problem complexity: NP-Complete problems get higher weights
- Energy efficiency: lower energy per operation = higher score
- Solution quality: exact solutions vs approximations

B.2 Dynamic Adaptation

- Block time emerges from verification performance
- Difficulty adjusts to maintain work score distribution
- Deflation occurs naturally as cumulative work grows
- Tier distribution balances through market forces

C User Submissions Integration

- Computational proof: Subset Sum (NP-Complete) provides verifiable asymmetry
- Practical utility: User submissions system allows solving real problems
- Aggregation strategies: ANY (first solution), BEST (quality competition), MULTIPLE (diverse solutions), STATISTICAL (sample collection)
- Bounty system: Users pay for solutions, miners earn from both block rewards and bounties

D Mathematical Proof: Exponential Dimensional Tokenomics

This appendix presents the rigorous mathematical framework underlying COINjecture's consensus and tokenomics design. The complete proof is available in the companion paper "Exponential Dimensional Tokenomics: A Mathematical Framework for Multi-Scale Cryptocurrency Stability."

D.1 Critical Complex Equilibrium

We model decentralized consensus as a coupled oscillator system with complex-valued state.

Definition 1 (Consensus State Dynamics)**.** The network consensus state $\psi(t) \in \mathbb{C}$ evolves according to:

$$\frac{d\psi}{dt} = \mu\psi(t) \tag{2}$$

where the complex eigenvalue is:

$$\mu = -\eta + i\lambda \tag{3}$$

with:

- $\eta > 0$: damping ratio (dissipation rate)
- $\lambda \geq 0$: coupling strength (synchronization rate)

Definition 2 (Bounded Dynamics)**.** For stability without unbounded growth or decay:

$$|\mu|^2 = \eta^2 + \lambda^2 = 1 \tag{4}$$

Geometric Interpretation: The eigenvalue μ lies on the unit circle in the complex plane.

Definition 3 (Critical Complex Equilibrium)**.** The system achieves critical equilibrium when real and imaginary components have equal magnitude:

$$|\mathrm{Re}(\mu)| = |\mathrm{Im}(\mu)| \implies \eta = \lambda \tag{5}$$

D.2 The Satoshi Constant

Theorem 1 (The Satoshi Constant)**.** *At critical equilibrium under the unit circle constraint, the unique solution is:*

$$\boxed{\eta = \lambda = \frac{1}{\sqrt{2}} \approx 0.7071} \tag{6}$$

Proof. Substituting $\eta = \lambda$ from (5) into (4):

$$\lambda^2 + \lambda^2 = 1 \tag{7}$$

$$2\lambda^2 = 1 \tag{8}$$

$$\lambda^2 = \frac{1}{2} \tag{9}$$

$$\lambda = \frac{1}{\sqrt{2}} \quad \text{(taking the positive root since } \lambda \geq 0\text{)} \tag{10}$$

Therefore $\eta = \lambda = \frac{1}{\sqrt{2}}$. □

This represents **critical damping** in control theory: the fastest possible convergence to consensus without oscillatory overshoot.

D.3 Exponential Dimensional Scales

Definition 4 (Dimensional Economic Scales)**.** Define dimensional scales as magnitude projections at specific dimensionless time points $\{\tau_n\}_{n=1}^{N}$:

$$D_n = e^{-\eta\tau_n} = e^{-\tau_n/\sqrt{2}} \tag{11}$$

where $\tau_n = t_n/\tau_c$ is dimensionless time measured in units of the consensus time constant $\tau_c = 1/\eta = \sqrt{2}$.

Scale	**Time** τ_n	**Value** D_n	**Description**
D_1	0.00	1.000	Genesis scale
D_2	0.20	0.867	Coupling scale
D_3	0.41	0.750	First harmonic
D_4	0.68	0.618	Golden ratio scale (φ^{-1})
D_5	0.98	0.500	Half-scale (2^{-1})
D_6	1.36	0.382	Second golden scale (φ^{-2})
D_7	1.96	0.250	Quarter-scale (2^{-2})
D_8	2.72	0.146	Euler scale

Table 1: The Eight Dimensional Scales

D.4 Emergence of Fundamental Constants

Remarkably, fundamental mathematical constants appear naturally in the dimensional scale structure:

Golden Ratio: $D_4 = \varphi^{-1} = \frac{\sqrt{5}-1}{2} \approx 0.618$ emerges at $\tau_4 \approx 0.68$
Powers of Two: $D_5 = 2^{-1} = 0.500$ and $D_7 = 2^{-2} = 0.250$ emerge exactly
Euler's Number: D_8 appears at $\tau_8 \approx e \approx 2.718$
These constants emerge endogenously from the exponential decay structure, not by design.

D.5 The Viviani Oracle: Performance Envelope

We introduce a geometric oracle based on Viviani's theorem that defines the system's performance envelope.

Theorem 2 (Viviani's Theorem)**.** *For any point inside an equilateral triangle, the sum of the perpendicular distances from the point to the three sides is constant and equal to the altitude of the triangle.*

Definition 5 (Oracle Metric)**.** For any point (η, λ) in the consensus state space, define the oracle metric:

$$\Delta(\eta, \lambda) = \frac{d_1(\eta, \lambda) + d_2(\eta, \lambda) + d_3(\eta, \lambda)}{\sqrt{3}/2} - 1 \tag{12}$$

where d_i are the perpendicular distances to the three sides of the consensus triangle.

Theorem 3 (Critical Equilibrium Oracle Reading)**.** *At the critical equilibrium point* $(\eta, \lambda) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$*, the oracle metric is:*

$$\Delta\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = 0.231 \tag{13}$$

representing 23.1% operation above the conservative stability bound.

Performance Regimes:

- **Idle** ($\Delta < 0.1$): Conservative, slow convergence
- **Cruise** ($0.1 \leq \Delta < 0.2$): Moderate performance, stable
- **Performance** ($0.2 \leq \Delta < 0.3$): Optimal convergence regime (Critical equilibrium here)
- **Redline** ($\Delta \geq 0.3$): Fast but oscillatory, requires active control

The oracle metric $\Delta = 0.231$ is:

- **Dimensionless**: A pure geometric ratio
- **Self-referenced**: Measured against the consensus triangle
- **Empirically grounded**: Derivable from observed network dynamics
- **Geometrically invariant**: Independent of parameter scaling

D.6 Financial Primitive Principles

All financial primitives in this framework adhere to three fundamental principles:

1. **Dimensionless**: No arbitrary limits or units. All measures are pure ratios or dimensionless quantities derived from network state.
2. **Self-Referenced**: Measured against network state (consensus magnitude $|\psi(t)|$, time constant τ_c, etc.) rather than external parameters.
3. **Empirically Grounded**: Derived from actual network behavior (consensus dynamics, measured decay rates) rather than assumptions or arbitrary parameters.

D.7 Tokenomics Implementation

Token Supply Allocation: For total supply $S_{\text{total}}(t)$ at time t, allocate across dimensions proportionally:

$$S_n(t) = \frac{\tilde{D}_n(t)}{\sum_{k=1}^{8} \tilde{D}_k(t)} \cdot S_{\text{total}}(t) = p_n(t) \cdot S_{\text{total}}(t) \tag{14}$$

where $\tilde{D}_n(t) = |\psi(t)| \cdot D_n$ is the dimensional scale measured against current consensus magnitude.

Unlock Schedule: Tokens in pool D_n unlock according to:

$$U_n(\tau) = \begin{cases} 0 & \text{if } \tau < \tau_n \\ 1 - e^{-\eta(\tau - \tau_n)} & \text{if } \tau \geq \tau_n \end{cases} \tag{15}$$

Yield Structure: Yield rate for pool D_n is derived from network's actual consensus dynamics:

$$r_n(\tau) = \frac{|\dot{\psi}(\tau)|}{|\psi(\tau)|} \cdot \frac{D_n}{D_1} = \eta \cdot e^{-\eta \tau_n} \tag{16}$$

All yields are dimensionless, self-referenced to network consensus dynamics, and empirically grounded in actual network behavior.

D.8 Stability Analysis

Theorem 4 (Global Asymptotic Stability)**.** *The equilibrium state $\psi = 0$ is globally asymptotically stable under the critical eigenvalue $\mu = -\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$.*

Proof. Using the Lyapunov function $V(\psi) = |\psi|^2$:

$$\frac{dV}{dt} = -2\eta|\psi|^2 = -\sqrt{2}|\psi|^2 < 0 \tag{17}$$

Since $\frac{dV}{dt} < 0$ for all $\psi \neq 0$, the system is globally asymptotically stable. □

Convergence Rate: The system converges to equilibrium exponentially with rate $\eta = \frac{1}{\sqrt{2}}$:

$$|\psi(t)| \leq |\psi_0| \cdot e^{-t/\sqrt{2}} \tag{18}$$

D.9 Conclusion

This framework provides the first rigorous mathematical unification of:

- **Consensus dynamics**: Complex eigenvalue evolution on unit circle
- **Tokenomics**: Multi-scale dimensional allocation
- **Stability**: Provable convergence guarantees

All aspects are governed by a single universal constant: $\frac{1}{\sqrt{2}}$.

The emergence of fundamental constants (φ, e, 2^n) from first principles suggests a deep connection between computational work, economic stability, and universal mathematics.

For the complete mathematical proof with all theorems, proofs, and detailed derivations, see the companion paper "Exponential Dimensional Tokenomics: A Mathematical Framework for Multi-Scale Cryptocurrency Stability."